Assignment: Polynomial Regression

1. Introduction

In this assignment, you will work with two personalised datasets and develop polynomial regression models to predict a continuous target variable y from the given input variables. The datasets are individually assigned to each student. Therefore, you must use the datasets corresponding to your own roll number and problem IDs.

The objective is to obtain accurate predictions on the provided test datasets using only polynomial regression.

2. Problem Statements

You are the Lead Renewable Energy Engineer managing a multi-stage geothermal power plant expansion project.

Phase 1: Power Plant Steam Turbine Optimization (var1)

Scenario: Before drilling new production wells, you must optimize the surface energy generation facility. Your plant operates a multi-stage steam turbine system. Due to slight variations in manufacturing tolerances and local water chemistry across regional plants, your specific plant setup behaves uniquely compared to your colleagues' facilities.

The turbine efficiency is governed by six key operational parameters $(x_1, x_2, x_3, x_4, x_5, x_6)$, representing percentage deviations:

x_1 High-pressure steam valve adjustment

x_2 Condenser coolant flow rate adjustment

x_3 Re-injection pump hydraulic pressure

x_4 Turbine blade pitch angle

x_5 Non-condensable gas exhaust valve rate

x_6 Steam inlet pressure adjustment

Based on these settings, the plant outputs a Net Power Score (y). The power score can be modeled using a polynomial of a moderate degree (up to degree 10) using the given parameters.

Your Task: Using historical plant calibration logs (`train`), build a polynomial regression model to predict the Net Power Score.

Phase 2: Subterranean Thermal Reservoir Mapping (var2)

Scenario: With your surface plant optimized, Phase 2 requires you to select the exact coordinates for drilling new geothermal extraction wells. You are assigned to map a specific, high-potential geological survey block.

You deploy seismic and thermal probe sensors across a 3D grid around your central basecamp. The variables x_1, x_2, and x_3 represent spatial coordinate offsets in meters from the site landmark:

- x_1 :East-West coordinate offset
- x_2 :North-South coordinate offset

- x_3 :Vertical depth offset relative to the basecamp

At each 3D coordinate, the sensor network records a Thermal Anomaly Score (y).

Because off complex geological structures, the heat map in your sector resembles a high-degree polynomial (up to degree 20).

Your Task: Due to survey budget limits, you have only collected physical core samples at a random scatter off 3D points (`train`). You must build a model to predict the Thermal Anomaly Score across the remaining planned drilling sites (`test`) to identify the exact coordinates and depth for drilling high-yield wells.

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2. Dataset Format

Each student will receive two **preprocessed and cleaned** datasets for two different problems. The files follow the naming convention:

<ROLLNO>_{train/test}_var<PRBID>.csv

For example,

```
BT2024001_train_var1.csv
BT2024001_test_var1.csv

BT2024001_train_var2.csv
BT2024001_test_var2.csv
```

where:

- `ROLLNO` denotes the student's roll number.
- `train` and `test` identify the training and testing datasets respectively.
- `PRBID` identifies the assigned problem.

The training and testing files corresponding to each problem must be treated as separate datasets. The target variable is `y`.

Download datasets here

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4. Evaluation Metrics

Your predictions on the `test` sets will be evaluated against the hidden ground truth. The primary metrics for evaluation will be:

- **Mean Squared Error (MSE):**To measure the average squared difference between your predicted values and the actual values.
- R^2 **Score (Coecient oDetermination):** To evaluate how well your polynomial model captures the variance off the target variable.

You must carefully select the optimal degree for your polynomial models to avoid underfitting or overfitting the uniquely generated data provided to you. A polynomial off degree x means for each term, the sum off the powers off features adds atmost upto x.

5. Deliverables

You must submit the following:

- **Report (PDF):** A concise write-up (maximum 4–5 pages) documenting your approach, the polynomial degree chosen for each problem, the rationale behind it, and any other techniques you used.
- **Prediction Files:** The two completed test CSV files (one per problem) containing your predicted values for `y`. Refer to the sample submission for the format.

  `<ROLLNO>_pred_var<PRBID>.csv`

- **Github Repo:** A github repo which contains all the code used to train the models, and find inference.